

Certified Ethical Hacker (CEH) v13 – Course Outline

Module 01: Introduction to Ethical Hacking

Overview of Ethical Hacking concepts and terminologies

Ethical hacker roles and responsibilities

Types of hackers and attack motivations

Phases of ethical hacking methodology

Cyber laws, ethics, and compliance requirements

Information security controls and frameworks

Common attack vectors and threat landscape

Penetration testing vs vulnerability assessment

How AI enhances hacking and security defense

Module 02: Footprinting and Reconnaissance

- Passive vs active reconnaissance techniques
- Open Source Intelligence (OSINT) gathering
- Domain, DNS, and IP footprinting
- Website and email footprinting methods
- Social media and people intelligence
- Competitive intelligence gathering
- Network footprinting techniques
- Countermeasures for reconnaissance
- AI-powered reconnaissance tools

Module 03: Scanning Networks

- Network scanning concepts and objectives

- TCP/UDP scanning techniques
- Port scanning and service identification
- Network mapping and topology discovery
- Vulnerability scanning fundamentals
- Firewall and IDS detection techniques
- Scanning tools and methodologies
- Scan result analysis and reporting

Module 04: Enumeration

- Enumeration concepts and attack surface analysis
- NetBIOS and SMB enumeration
- SNMP enumeration techniques
- LDAP and directory service enumeration
- User and group enumeration methods
- Network resource and share discovery
- Enumeration tools and scripts
- Countermeasures against enumeration
- AI-assisted vulnerability scanning techniques

Module 05: Vulnerability Analysis

- Vulnerability assessment lifecycle
- Types of vulnerabilities and exposures
- CVE, CVSS, and vulnerability scoring
- Automated vs manual vulnerability analysis
- Network and host-based vulnerabilities
- Application and configuration weaknesses

- Vulnerability scanning tools overview
- Risk analysis and remediation strategies

Module 06: System Hacking

- Password cracking techniques
- Authentication and authorization attacks
- Privilege escalation methods
- Maintaining access and persistence
- Covering tracks and log manipulation
- Rootkits and backdoor techniques
- System exploitation tools
- System hardening countermeasures
- AI-based malware detection & defense

Module 07: Malware Threats

- Types of malware and attack vectors
- Malware propagation techniques
- Trojans, viruses, worms, and ransomware
- Spyware, keyloggers, and botnets
- Malware analysis basics
- Anti-malware evasion techniques
- Malware detection and prevention
- Incident response for malware attacks
- Using AI to analyze and mitigate malware threats

Module 08: Sniffing

- Packet sniffing concepts and methods
- Active and passive sniffing techniques
- ARP poisoning and MAC spoofing
- Session hijacking attacks
- Credential sniffing methods
- Network protocol vulnerabilities
- Sniffing tools and analysis
- Sniffing countermeasures
- AI-driven anomaly detection in traffic analysis

Module 09: Social Engineering

- Social engineering attack lifecycle
- Human-based attack techniques
- Phishing, spear phishing, and whaling
- Pretexting, baiting, and tailgating
- Social engineering via email and phone
- Social media exploitation
- Detection and prevention strategies
- Security awareness best practices
- AI-driven phishing detection and prevention

Module 10: Denial-of-Service (DoS) Attacks

- DoS and DDoS attack concepts
- Volumetric, protocol, and application attacks
- Botnets and amplification attacks
- Distributed attack mechanisms

- Attack tools and simulation
- Impact analysis of DoS attacks
- Detection and mitigation techniques
- DoS prevention strategies
- AI-based anomaly detection for DDoS mitigation

Module 11: Session Hijacking

- Session management concepts
- Cookie and token-based attacks
- Session fixation techniques
- Man-in-the-Middle (MITM) attacks
- Web session hijacking methods
- Tools for session attacks
- Detection techniques
- Secure session handling practices

Module 12: Evading IDS, Firewalls, and Honeypots

- Intrusion detection and prevention systems
- Firewall architectures and limitations
- IDS/IPS evasion techniques
- Traffic fragmentation and obfuscation
- Tunneling and covert channels
- Honeypot detection methods
- Anti-forensic techniques
- Defensive countermeasures

Module 13: Hacking Web Servers

- Web server architecture and components
- Common web server vulnerabilities
- Misconfiguration and patching issues
- Directory traversal attacks
- Web server fingerprinting
- Attacking Apache, Nginx, IIS
- Web server attack tools
- Hardening and secure configuration

Module 14: Hacking Web Applications

- Web application architecture
- OWASP Top 10 vulnerabilities
- SQL Injection and XSS attacks
- Authentication and session flaws
- File upload and command injection
- API security weaknesses
- Web application testing tools
- Secure coding and protection
- AI-assisted security testing and automated scanning

Module 15: SQL Injection

- SQL injection fundamentals
- Types of SQL injection attacks
- Error-based and blind SQL injection
- Database enumeration techniques

- Exploiting authentication bypass
- SQL injection automation tools
- Detection and prevention techniques
- Secure database coding practices

Module 16: Hacking Wireless Networks

- Wireless networking standards
- Wireless encryption mechanisms
- WEP, WPA, WPA2, WPA3 attacks
- Rogue access point attacks
- Evil twin and MITM attacks
- Wireless sniffing and cracking tools
- Wireless penetration testing
- Wireless security best practices

Module 17: Hacking Mobile Platforms

- Mobile platform architectures
- Android and iOS security models
- Mobile malware threats
- Application sandboxing weaknesses
- Mobile network attacks
- Mobile app vulnerability testing
- Mobile hacking tools
- Mobile device security controls

Module 18: IoT and OT Hacking

- IoT and OT ecosystem overview
- Common IoT vulnerabilities
- Embedded system security issues
- Firmware and hardware attacks
- SCADA and industrial control systems
- IoT attack tools and techniques
- Securing IoT devices
- Risk mitigation strategies
- AI-driven security solutions for cloud environments

Module 19: Cloud Computing

- Cloud service models and deployment types
- Cloud security shared responsibility model
- Cloud-specific attack vectors
- Identity and access management flaws
- Container and virtualization risks
- Cloud penetration testing tools
- Cloud logging and monitoring
- Cloud security best practices

Module 20: Cryptography

- Cryptography fundamentals
- Symmetric and asymmetric encryption
- Hashing and digital signatures
- PKI and certificate management
- Cryptographic attacks

- Encryption implementation flaws
- Secure communication protocols
- Cryptography countermeasures